Doiraviy diagrammada ma'lumotlarni taqdim etish

Jadval va ustunli diagramma har bir miqdorni alohida ko'rsatadi. Butun narsa qanday qismlardan iboratligini bir qarashda ko'rish uchun esa doiraviy diagramma qulay.

1-QISM

Mavzu

ESLAB OLAYLIK

- Qism butunning qanday ulushi ekanini kasr ko'rsatadi: 30 ta daftardan 10 tasi — $\frac{10}{30} = \frac{1}{3}$ qismi.
- Kasrni foizga aylantirish uchun uni 100% ga ko'paytiramiz: $\frac{3}{4} \cdot 100\% = 75\%$.
- Doira teng ikkiga bo'linsa — yarim doira, teng to'rtga bo'linsa — chorak doira hosil bo'ladi. Chorak doiraning markazdagi burchagi — to'g'ri burchak.

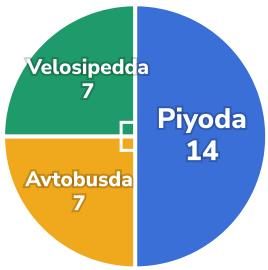
QOIDA · DOIRAVIY DIAGRAMMA

Doiraviy diagrammada doira bir butunni ifodalaydi. Barcha ulushlarning yig'indisi esa 1 ga teng.

1-MISOL · SONLAR BILAN

Sinfdagi 28 o'quvchidan maktabga qanday kelishini so'rashdi.

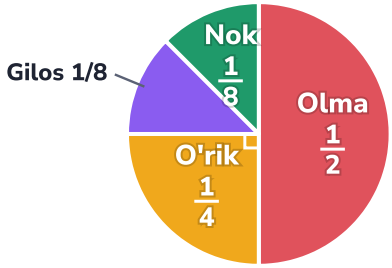
Qanday keladi	O'quvchilar soni
Piyoda	14
Avtobusda	7
Velosipedda	7



1. 14 — 28 ning yarmi. Piyoda keluvchilar — yarim doira.
2. Avtobusda va velosipedda keluvchilar soni teng (7 tadan). Ularning bo'laklari ham teng — har biri chorak doira.

2-MISOL · ULUSHLAR BILAN

Bog'dagi daraxtlarning $\frac{1}{2}$ qismi olma, $\frac{1}{4}$ qismi o'rik, $\frac{1}{8}$ qismi gilos, $\frac{1}{8}$ qismi nok.



1. Olma — yarim doira, o'rik — chorak doira. Gilos va nok — chorakning yarmidan.
2. Tekshiramiz: $\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{8}=\frac{4}{8}+\frac{2}{8}+\frac{1}{8}+\frac{1}{8}=1$.



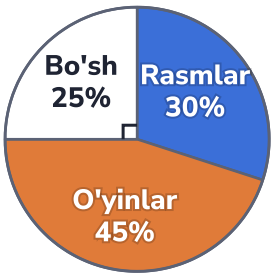
Bo'lak qancha katta bo'lsa, butunning shuncha katta qismini ko'rsatadi. Teng sonlarga teng bo'laklar to'g'ri keladi.

QOIDA · FOIZLAR BILAN

Doira 100% ni ifodalaydi.

3-MISOL · QOLGAN QISM

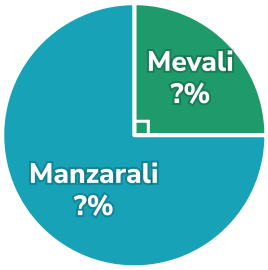
Telefon xotirasining 30% ini rasmlar, 45% ini o'yinlar egallagan. Qolgan qismi bo'sh.



1. Butun xotira — 100%. Bo'sh qism: $100\% - 30\% - 45\% = 25\%$.
2. 25% — butunning $\frac{1}{4}$ qismi, shuning uchun bo'sh qism — chorak doira.

4-MISOL · TO'G'RI BURCHAK BELGISI

Maktab hovlisiga mevali va manzarali ko'chatlar ekildi. Diagrammada mevali ko'chatlar bo'lagida to'g'ri burchak belgisi bor. Har bir bo'lak necha foiz?



1. To'g'ri burchakli bo'lak — chorak doira: $\frac{1}{4} \cdot 100\% = 25\%$. Mevali ko'chatlar — 25%.
2. Manzarali ko'chatlar: $100\% - 25\% = 75\%$.

USUL · SONLAR, KASRLAR VA FOIZLAR

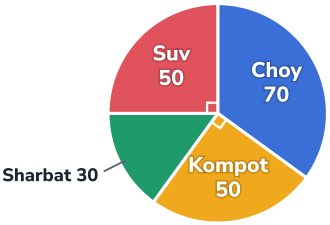
1. Jamini topamiz.
2. Kasr ko'rinishida: $\frac{\text{soni}}{\text{jami}}$, so'ng kasrni qisqartiramiz.
3. Foiz ko'rinishida: $\frac{\text{soni}}{\text{jami}} \cdot 100\%$.

Xuddi shu ma'lumotlarni doiraviy diagramma orqali sonlar, kasrlar yoki foizlarda ifodalash mumkin.

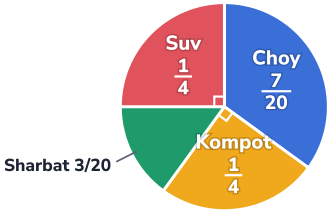
5-MISOL · BITTA MA'LUMOT — UCH XIL YOZUV

Maktab oshxonasida bir tanaffusda 200 ta ichimlik olindi.

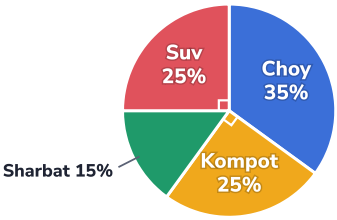
	Choy	Kompot	Sharbat	Suv	Jami
Soni	70	50	30	50	200
Kasr ko'rinishida	$\frac{70}{200} = \frac{7}{20}$	$\frac{50}{200} = \frac{1}{4}$	$\frac{30}{200} = \frac{3}{20}$	$\frac{50}{200} = \frac{1}{4}$	1
Foiz ko'rinishida	$\frac{7}{20} \cdot 100\% = 35\%$	$\frac{1}{4} \cdot 100\% = 25\%$	$\frac{3}{20} \cdot 100\% = 15\%$	$\frac{1}{4} \cdot 100\% = 25\%$	100%



Sonlar



Kasrlar



Foizlar

Tekshiramiz: $35\% + 25\% + 15\% + 25\% = 100\%$.



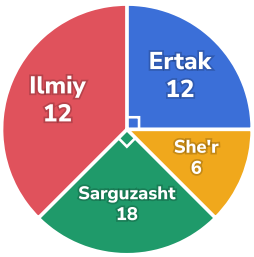
Uchala diagrammada bo'laklar bir xil. Faqat yozuvlari boshqa: son, kasr yoki foiz.

2-QISM

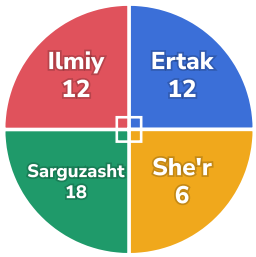
Misollar ishlash

1-MISOL · QAYSI DIAGRAMMA TO'G'RI?

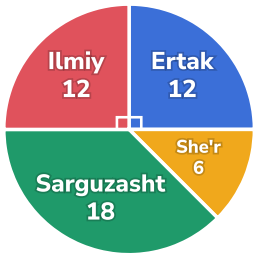
Maktab kutubxonasidan bir haftada 48 ta kitob olindi: 12 ta ertak, 6 ta she'riy kitob, 18 ta sarguzasht va 12 ta ilmiy kitob. Qaysi diagramma bu ma'lumotni to'g'ri ifodalagan?



A



B

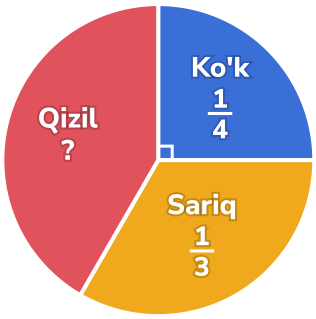


C

1. Har bir ulushni topamiz: ertak va ilmiy — $\frac{12}{48} = \frac{1}{4}$, she'r — $\frac{6}{48} = \frac{1}{8}$, sarguzasht — $\frac{18}{48} = \frac{3}{8}$.
2. Demak ertak va ilmiy kitoblar — teng chorak doiralar, sarguzasht — eng katta bo'lak, she'r — eng kichik bo'lak.
3. A da ilmiy kitoblar bo'lagi sarguzashtnikidan katta — xato. B da hamma bo'laklar teng, lekin 6 va 18 teng emas — xato. To'g'ri javob: C.

2-MISOL · QOLGAN QISMNING ULUSHI

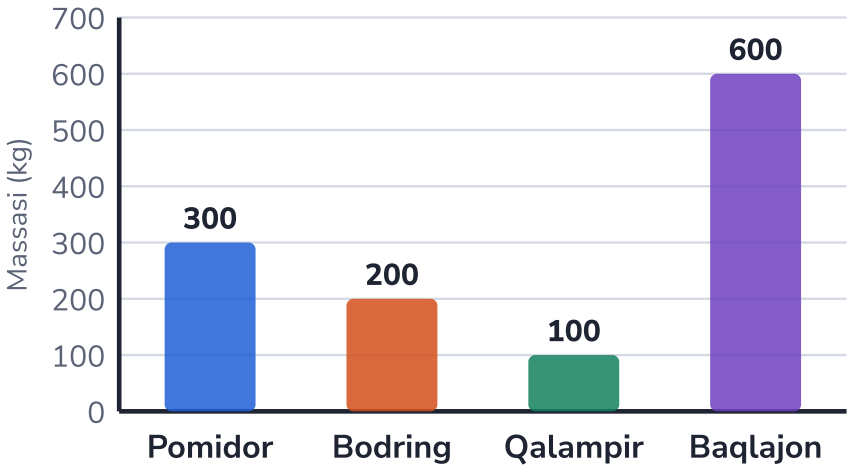
Do'kondagi havo sharlarining $\frac{1}{4}$ qismi ko'k, $\frac{1}{3}$ qismi sariq, qolganlari qizil. Doiraviy diagrammada qizil sharlar bo'lagiga qanday kasr yoziladi? Qaysi bo'lak eng katta bo'ladi?

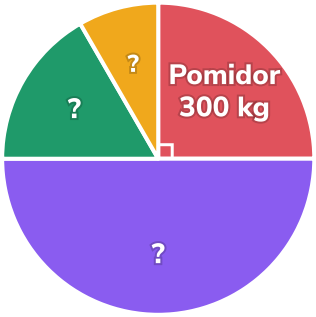


1. Hamma sharlar — 1. Kasrlarni 12 maxrajga keltiramiz: $\frac{1}{4} = \frac{3}{12}, \frac{1}{3} = \frac{4}{12}$.
2. Qizil sharlar: $1 - \frac{3}{12} - \frac{4}{12} = \frac{12}{12} - \frac{7}{12} = \frac{5}{12}$.
3. $\frac{5}{12}$ — eng katta ulush, shuning uchun eng katta bo'lak — qizil sharlarniki.

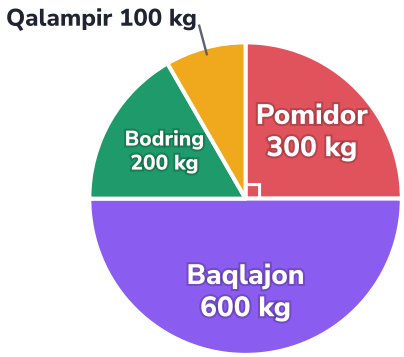
3-MISOL · USTUNLI DIAGRAMMADAN DOIRAVIY DIAGRAMMAGA

Ustunli diagrammada issiqxonadan bir kunda uzilgan hosil ko'rsatilgan. Doiraviy diagrammaning bo'sh bo'laklariga nom va massani yozing.



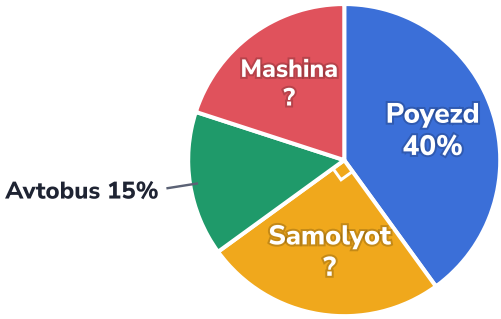


1. Jami: $300 + 200 + 100 + 600 = 1200$ kg.
2. Ulushlar: pomidor $\frac{300}{1200} = \frac{1}{4}$, bodring $\frac{200}{1200} = \frac{1}{6}$, qalampir $\frac{100}{1200} = \frac{1}{12}$, baqlajon $\frac{600}{1200} = \frac{1}{2}$.
3. Yarim doira — baqlajon (600 kg). Qolgan ikki bo'lakdan kattasi — bodring (200 kg), eng kichigi — qalampir (100 kg).
4. Tekshiramiz: pomidor — chorak doira, qolgan uchtasi birga — $\frac{3}{4}$ qism: $600 + 200 + 100 = 900$ kg. Chorak doira: $900 : 3 = 300$ kg — pomidor massasiga teng.



4-MISOL · FOIZLARDAN KASRLARGA

Diagrammada sayyohlar shaharga qanday kelgani foizlarda ko'rsatilgan. Jadvalni to'ldiring.



Transport	Foizi	Qismi
Poyezd	40%	$\frac{40}{100} = \frac{2}{5}$
Samolyot	25%	$\frac{1}{4}$
Avtobus	15%	$\frac{15}{100} = \frac{3}{20}$
Mashina	20%	$\frac{20}{100} = \frac{1}{5}$

1. Samolyot bo'lagida to'g'ri burchak belgisi bor — chorak doira, ya'ni 25%.
2. Mashina: $100\% - 40\% - 25\% - 15\% = 20\%$.
3. Foizni kasrga aylantiramiz: foizni 100 ga bo'lib, kasrni qisqartiramiz. Masalan $15\% = \frac{15}{100} = \frac{3}{20}$.

Diagramma tayyormi? Ulushlarni qo'shib tekshiring: kasrlar yig'indisi 1, foizlar yig'indisi 100% chiqishi kerak.



DIQQAT!

1. Kasr — soni : jami. 48 ta kitobdan 12 tasi ertak: ✗ $\frac{12}{36}$ (qolganlarga bo'lingan)
   ✓ $\frac{12}{48} = \frac{1}{4}$
2. Sonni foiz deb yozmang. 80 o'quvchidan 20 tasi: ✗ 20% ✓ $\frac{20}{80} \cdot 100\% = 25\%$
3. Kasrlarni qo'shishda maxrajlarni qo'shmang: ✗ $1 - \frac{1}{4} - \frac{1}{3} = 1 - \frac{2}{7} = \frac{5}{7}$
   ✓ $1 - \frac{3}{12} - \frac{4}{12} = \frac{5}{12}$
4. Teng sonlarga teng bo'laklar, katta songa katta bo'lak to'g'ri keladi. 6 ta va 18 ta bir xil bo'lak bilan ko'rsatilmaydi.

3-QISM

Mustahkamlash

1. Jadvalda bog'dan ikki kunda uzilgan mevalar (kg) ko'rsatilgan.

	1-kun	2-kun	Jami
Olma	48	?	110
Nok	?	35	75
Behi	27	33	?
Jami	?	?	?

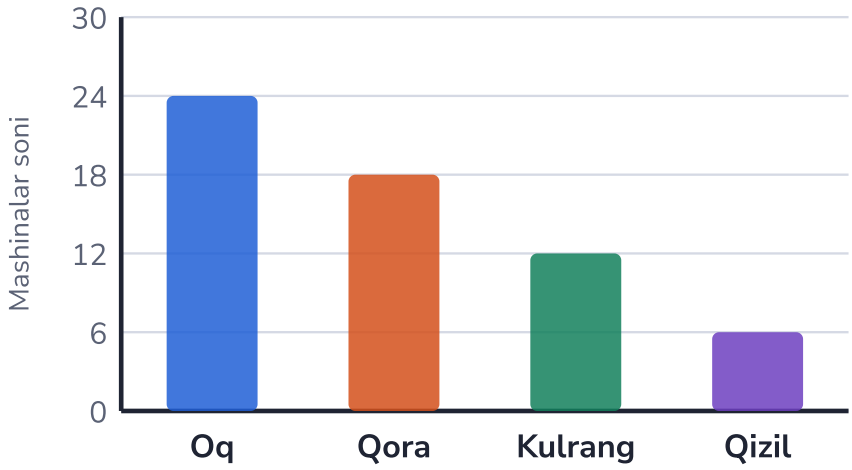
a) Jadvalni daftaringizga chizib to'ldiring.

b) Qaysi kuni ko'proq meva uzildi va qancha ko'p?

c) 2-kuni nok olmadan necha kg kam uzildi?

d) Ikki kunda uzilgan barcha mevalarning qanday qismi behi? Javobni qisqarmas kasr bilan yozing.

2. Ustunli diagrammada avtoturargohdagi mashinalar rangi ko'rsatilgan.



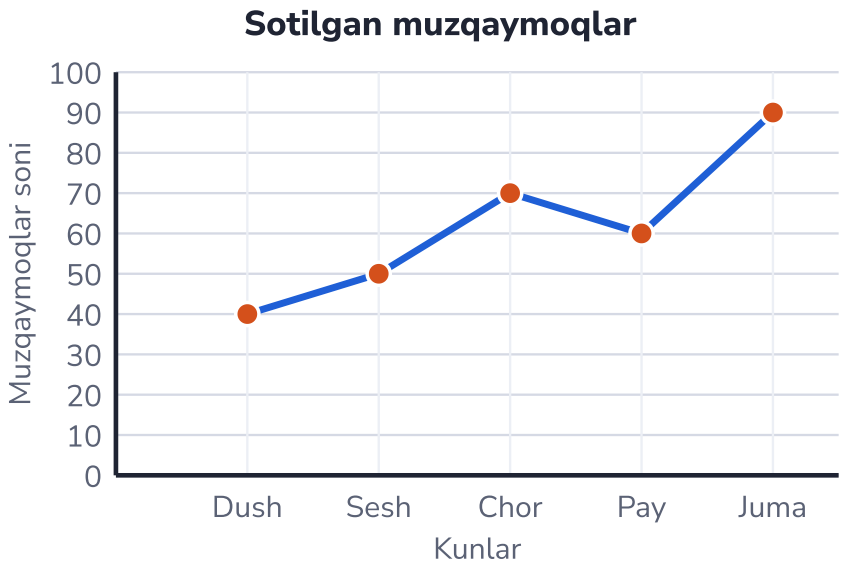
a) Avtoturargohda jami nechta mashina bor?

b) Oq mashinalar qizil mashinalardan necha marta ko'p?

c) Har bir mashina bir soat uchun 3000 so'm to'laydi. Kulrang mashinalar bir soatda jami qancha to'laydi?

d) Mashinalarning qanday qismi qora? Javobni qisqarmas kasr bilan yozing.

3. Chiziqli diagrammada do'konda 5 kun davomida sotilgan muzqaymoqlar soni ko'rsatilgan.



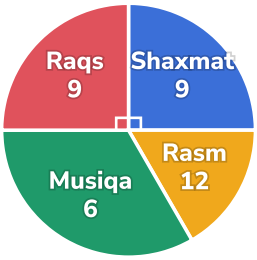
a) Chorshanba kuni nechta muzqaymoq sotildi?

b) Qaysi kuni sotuv oldingi kundagidan kamaydi?

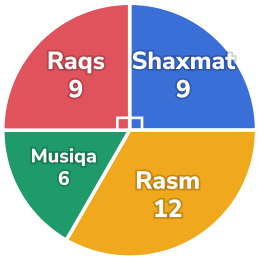
c) 5 kunda jami nechta muzqaymoq sotildi?

d) Bir kunda o'rtacha nechta muzqaymoq sotilgan?

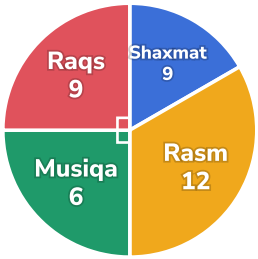
4. To'garakka qatnaydigan 36 o'quvchidan 9 tasi shaxmat, 12 tasi rasm, 6 tasi musiqa, 9 tasi raqs bilan shug'ullanadi. Qaysi diagramma bu ma'lumotga mos? Qolgan ikkitasida qanday xato bor?



A



B



C

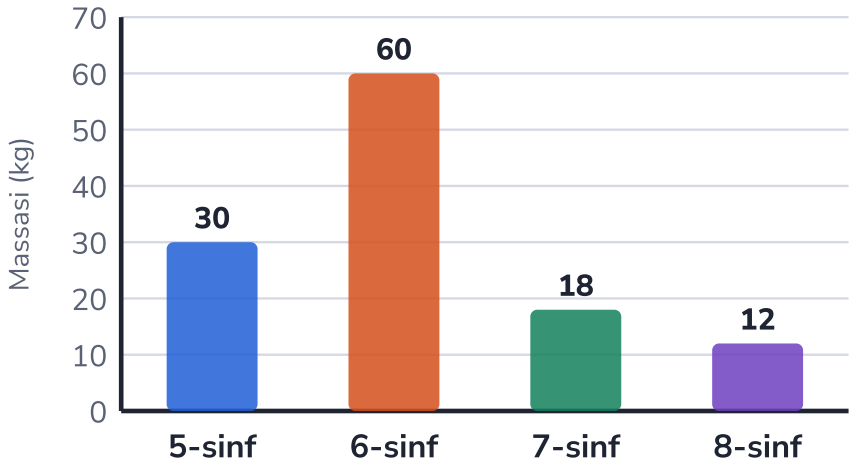
5. Akvariumdagi baliqlarning $\frac{3}{8}$ qismi qizil, $\frac{1}{4}$ qismi sariq, qolganlari kulrang.
   a) Kulrang baliqlar baliqlarning qanday qismini tashkil etadi?
   b) Doiraviy diagrammada qaysi ikki bo'lak teng bo'ladi?
   c) Doirani 8 ta teng bo'lakka bo'lib, diagrammani daftaringizga chizing.

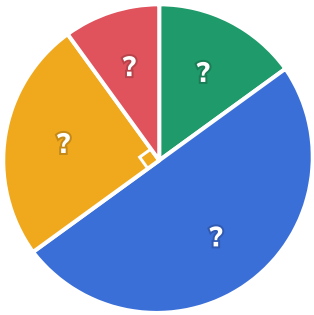
6. Sayyohlar bosib o'tgan yo'lning 40% i asfalt yo'l, 35% i tuproq yo'l, qolgani tog' so'qmog'i.
   a) Tog' so'qmog'i yo'lning necha foizi?
   b) Doiraviy diagrammada qaysi bo'lakka to'g'ri burchak belgisi qo'yiladi?

7. Sport maydonchasida bir hafta davomida 60 o'quvchi mashg'ulot qildi: 24 tasi futbol, 18 tasi voleybol, 18 tasi basketbol o'ynadi. Jadvalni to'ldiring. Doiraviy diagrammada qaysi ikki bo'lak teng bo'ladi?

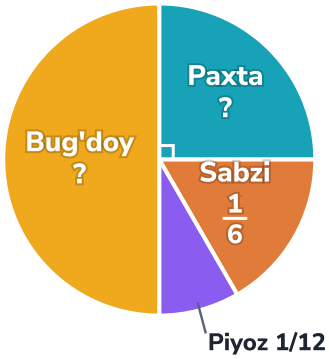
	Futbol	**Voleybol**	**Basketbol**	**Jami**
Soni	24	18	18	60
Kasr ko'rinishida	?	?	?	1
Foiz ko'rinishida	?	?	?	100%

8. Ustunli diagrammada maktab sinflari yiqqan qog'oz chiqindisi ko'rsatilgan. Doiraviy diagrammaning bo'laklariga sinf va massani yozing.





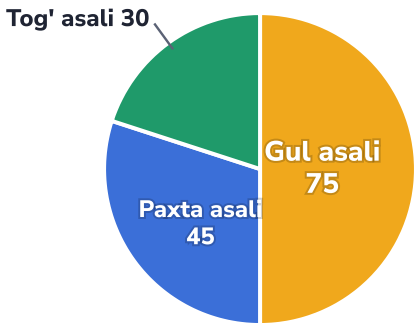
9. Doiraviy diagrammada fermer dalasidagi ekinlar ko'rsatilgan.



a) Bo'sh qolgan kasrlarni toping.

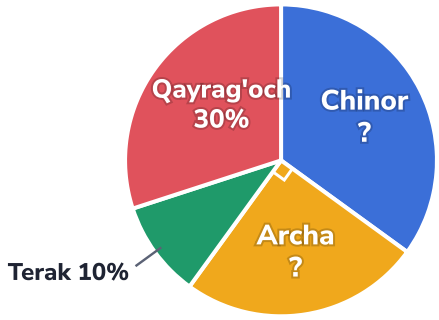
b) Qaysi ikki ekin birgalikda paxta bilan teng joy egallagan?

10. Doiraviy diagrammada asalarichi bir mavsumda yig'gan asal massasi (kg) ko'rsatilgan. Jadvalni to'ldiring.



Asal turi	Qismi	Foizi
Gul asali	?	?
Paxta asali	?	?
Tog' asali	?	?

11. Doiraviy diagrammada shahar parkidagi daraxtlar foizlarda ko'rsatilgan. Jadvalni to'ldiring.

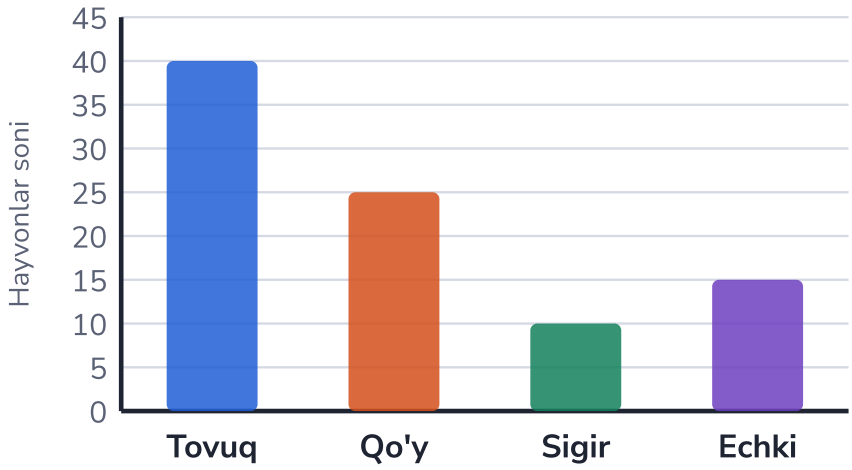


Daraxt	Qismi	Foizi
Chinor	?	?
Archa	?	?
Terak	?	10%
Qayrag'och	?	30%

4-QISM

Mustaqil ish

1. Ustunli diagrammada qishloqdagi bir hovlida boqiladigan hayvonlar soni ko'rsatilgan.

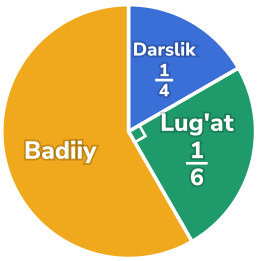


   a) Hovlida jami nechta hayvon bor?

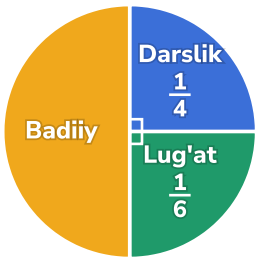
   b) Qo'ylar echkilardan nechta ko'p?

   c) Hayvonlarning qanday qismi tovuq? Javobni qisqarmas kasr bilan yozing.

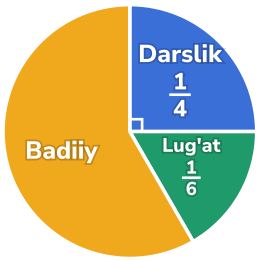
2. Kutubxona javonidagi kitoblarning $\frac{1}{4}$ qismi darslik, $\frac{1}{6}$ qismi lug'at, qolgani badiiy kitob. Qaysi diagramma to'g'ri chizilgan? Boshqalarida nima xato?



A



B



C

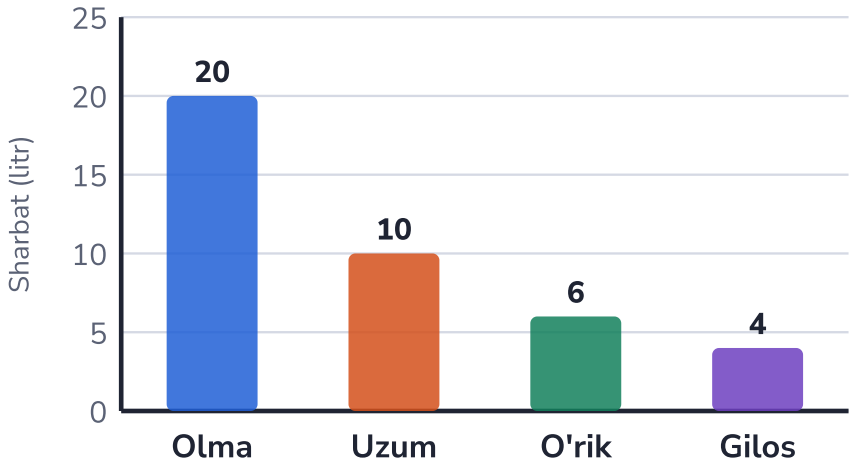
3. Shahar bog'iga bir kunda 200 kishi keldi: 120 tasi katta yoshlilar, 60 tasi bolalar, 20 tasi o'smirlar. Har bir guruhning ulushini kasr va foiz ko'rinishida yozing. Doiraviy diagrammada qaysi guruhning bo'lagi yarim doiradan katta bo'ladi?

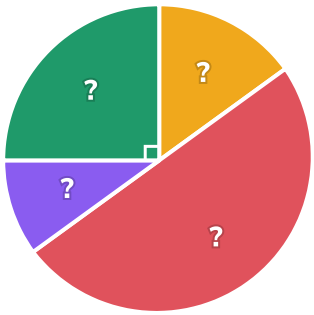
4. Qishloq yerlarining 45% i ekin maydoni, 30% i yaylov, qolgani bog'.

   a) Bog'lar yerning necha foizini egallaydi?

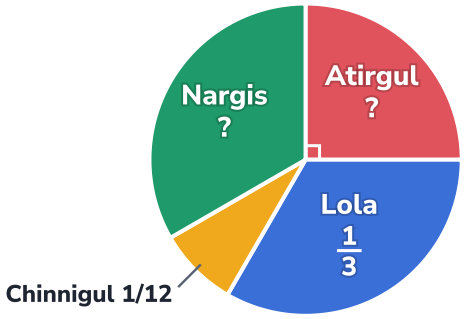
   b) Doiraviy diagrammada qaysi bo'lakka to'g'ri burchak belgisi qo'yiladi?

5. Ustunli diagrammada do'konda bir kunda sotilgan sharbat (litr) ko'rsatilgan. Doiraviy diagrammaning bo'laklariga sharbat nomi va kasrini yozing.





6. Doiraviy diagrammada guldastadagi gullar ko'rsatilgan.



a) Bo'sh qolgan kasrlarni toping.
b) Atirgul va chinnigul birgalikda guldastaning qanday qismini tashkil etadi?

7. **Xatoni toping.** Sport to'garagidagi 60 o'quvchidan 15 tasi suzish bilan shug'ullanadi. Sardor doiraviy diagrammaga «Suzish — 15%» deb yozdi. Xato nimada? To'g'ri foiz qancha?

8. **O'ylab ko'ring.** Doiraviy diagramma uchta bo'lakdan iborat. Birinchi bo'lak ikkinchisidan 2 marta katta. Uchinchi bo'lakda to'g'ri burchak belgisi bor. Har bir bo'lak necha foiz?

5-QISM

Javoblar

Mustahkamlash

1-masala

a) Olma 2-kuni: $110 - 48 = 62$; nok 1-kuni: $75 - 35 = 40$; behi jami: $27 + 33 = 60$. 1-kun jami: $48 + 40 + 27 = 115$, 2-kun jami: $62 + 35 + 33 = 130$, hammasi: 245 kg. b) 2-kuni, $130 - 115 = 15$ kg ko'p. c) $62 - 35 = 27$ kg kam. d) $\frac{60}{245} = \frac{12}{49}$.

2-masala

a) $24 + 18 + 12 + 6 = 60$ ta. b) $24 : 6 = 4$ marta. c) $12 \cdot 3000 = 36\,000$ so'm. d) $\frac{18}{60} = \frac{3}{10}$.

3-masala

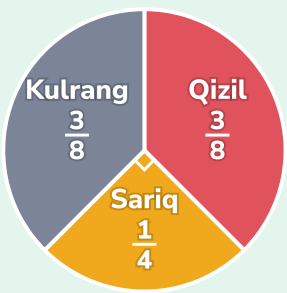
a) 70 ta. b) Payshanba (70 tadan 60 taga kamaydi). c) $40 + 50 + 70 + 60 + 90 = 310$ ta. d) $310 : 5 = 62$ ta.

4-masala

B. Shaxmat va raqs — $\frac{9}{36} = \frac{1}{4}$ dan (teng chorak doiralar), rasm — $\frac{12}{36} = \frac{1}{3}$ (eng katta bo'lak), musiqa — $\frac{6}{36} = \frac{1}{6}$ (eng kichik bo'lak). A da musiqa bo'lagi rasmnikidan katta, C da shaxmat va raqs bo'laklari teng emas.

5-masala

a) $1 - \frac{3}{8} - \frac{2}{8} = \frac{3}{8}$. b) Qizil va kulrang baliqlar ($\frac{3}{8}$ dan). c) Qizil — 3 bo'lak, sariq — 2 bo'lak, kulrang — 3 bo'lak:

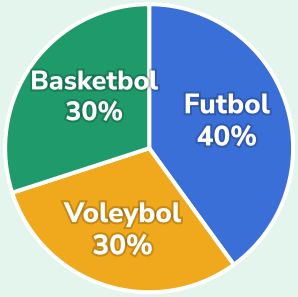


6-masala

a) $100\% - 40\% - 35\% = 25\%$. b) Tog' so'qmog'i bo'lagiga (25% — chorak doira).

7-masala

Futbol: $\frac{24}{60} = \frac{2}{5}$, 40%; voleybol: $\frac{18}{60} = \frac{3}{10}$, 30%; basketbol: $\frac{3}{10}$, 30%. Voleybol va basketbol bo'laklari teng.



8-masala

Jami: $30 + 60 + 18 + 12 = 120$ kg. Yarim doira — 6-sinf (60 kg), chorak doira — 5-sinf (30 kg). Qolgan ikkitasidan kattasi — 7-sinf (18 kg), kichigi — 8-sinf (12 kg). Tepadan soat mili bo'yicha: 7-sinf, 6-sinf, 5-sinf, 8-sinf.

9-masala

a) Paxta — to'g'ri burchak belgisi: $\frac{1}{4}$. Bug'doy: $1 - \frac{1}{4} - \frac{1}{6} - \frac{1}{12} = \frac{12}{12} - \frac{3}{12} - \frac{2}{12} - \frac{1}{12} = \frac{6}{12} = \frac{1}{2}$. b) Sabzi va piyoz: $\frac{2}{12} + \frac{1}{12} = \frac{3}{12} = \frac{1}{4}$ — paxta bilan teng.

10-masala

Jami: $75 + 45 + 30 = 150$ kg. Gul asali $\frac{75}{150} = \frac{1}{2}$, 50%; paxta asali $\frac{45}{150} = \frac{3}{10}$, 30%; tog' asali $\frac{30}{150} = \frac{1}{5}$, 20%.

11-masala

Archa — chorak doira: 25%, $\frac{1}{4}$. Chinor: $100\% - 25\% - 10\% - 30\% = 35\%$, $\frac{35}{100} = \frac{7}{20}$. Terak $\frac{10}{100} = \frac{1}{10}$, qayrag'och $\frac{30}{100} = \frac{3}{10}$.

Mustaqil ish

1-masala

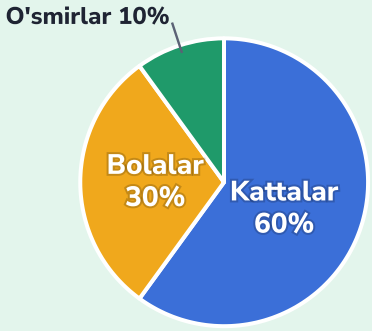
a) $40 + 25 + 10 + 15 = 90$ ta. b) $25 - 15 = 10$ ta. c) $\frac{40}{90} = \frac{4}{9}$.

2-masala

Badiiy kitoblar: $1 - \frac{3}{12} - \frac{2}{12} = \frac{7}{12}$ — yarim doiradan katta. Darslik — chorak doira, lug'at undan kichik. To'g'ri javob: C. A da darslik va lug'at bo'laklari almashib qolgan, B da lug'at ham chorak doira qilib chizilgan.

3-masala

Katta yoshlilar $\frac{120}{200} = \frac{3}{5}$, 60%; bolalar $\frac{60}{200} = \frac{3}{10}$, 30%; o'smirlar $\frac{20}{200} = \frac{1}{10}$, 10%. Katta yoshlilar bo'lagi yarim doiradan katta (60% > 50%).



4-masala

a) $100\% - 45\% - 30\% = 25\%$. b) Bog' bo'lagiga.

5-masala

Jami: $20 + 10 + 6 + 4 = 40$ litr. Olma $\frac{20}{40} = \frac{1}{2}$ (yarim doira), uzum $\frac{1}{4}$ (chorak doira), o'rik $\frac{6}{40} = \frac{3}{20}$, gilos $\frac{4}{40} = \frac{1}{10}$ (eng kichik). Tepadan soat mili bo'yicha: o'rik, olma, gilos, uzum.

6-masala

a) Atirgul — chorak doira: $\frac{1}{4}$. Nargis: $1 - \frac{3}{12} - \frac{4}{12} - \frac{1}{12} = \frac{4}{12} = \frac{1}{3}$. b) $\frac{3}{12} + \frac{1}{12} = \frac{4}{12} = \frac{1}{3}$ — lola yoki nargis bo'lagiday.

7-masala

Sardor sonni foiz deb yozgan. To'g'risi: $\frac{15}{60} = \frac{1}{4}$, $\frac{1}{4} \cdot 100\% = 25\%$.

8-masala

Uchinchi bo'lak — 25%. Qolgan ikkitasi birga: $100\% - 25\% = 75\%$. Ikkinchi bo'lakni 1 qism desak, birinchisi 2 qism, jami 3 qism: $75\% : 3 = 25\%$. Birinchi bo'lak 50%, ikkinchisi 25%, uchinchisi 25%.